Inference lab using the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Run a closed-form power calculation for a two-sample *t*-test with `pwr`.
- Simulate power for a scenario where closed-form solutions do not exist using `simr`.
- Describe the inputs to any power calculation (effect, variance, alpha, design).

Prerequisites

inference and mixed-model basics (we will fit a small `lmer`).

Background

Power is the probability of rejecting the null hypothesis when a specific alternative is true. Four inputs set it: the effect size you care about, the variability of the outcome, the significance level, and the sample size (with the design). Closed-form formulas exist for standard one- and two-sample tests and for some simple regression settings. For anything more elaborate — clustered data, non-linear models, interim looks, complex survival designs — you simulate.

Simulation-based power is conceptually straightforward: for a given scenario, generate many datasets under the alternative, run the planned analysis on each, and compute the fraction of times you reject the null. The `simr` package does this for models fit with `lme4`, allowing you to vary the sample size or the effect and trace out a power curve.

A useful rule of thumb: if your closed-form calculation is for a model simpler than the one you will actually fit, the closed-form is giving you a lower bound on the sample size you need, not an answer.

Setup

```r
library(tidyverse)
library(pwr)
library(lme4)
library(lmerTest)
library(simr)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

Detect a standardised mean difference of $d = 0.4$ between two groups with 80% power at a 5% two-sided significance level. Then power a mixed model with 20 clusters and 10 observations each to detect a fixed effect of 0.3.

2. Visualise

```r
pw <- map_dbl(ns, \(n) pwr.t.test(
  n = n / 2, d = 0.4, sig.level = 0.05, type = "two.sample"
)$power)

tibble(n = ns, power = pw) |>
  ggplot(aes(n, power)) +
  geom_line() +
  geom_hline(yintercept = 0.8, linetype = "dashed", colour = "grey40") +
  labs(x = "Total sample size", y = "Power")
```

3. Assumptions

Closed-form assumes equal variance across arms, independent observations, and the nominal effect size. The simulation below adds random intercepts at the cluster level and therefore relaxes independence.

4. Conduct

```r
res <- pwr.t.test(d = 0.4, power = 0.8, sig.level = 0.05, type = "two.sample")
res
```

```r
J   <- 20                              # clusters
nj  <- 10                              # obs per cluster
dat <- tibble(
  cluster = factor(rep(seq_len(J), each = nj)),
  arm     = rep(rep(c(0, 1), each = nj), length.out = J * nj),
  y       = rnorm(J * nj)              # placeholder; simr overwrites
)

m0 <- makeLmer(
  y ~ arm + (1 | cluster),
  fixef   = c(0, 0.3),
  VarCorr = 0.5,
  sigma   = 1.0,
  data    = dat
)
m0
```

```r
ps <- powerSim(m0, nsim = 50, test = fixed("arm"), progress = FALSE)
ps
```

Fifty simulations is enough for a teaching demo; in practice use at least 1000 for a stable estimate.

5. Concluding statement

> A two-sample t-test targeting $d = 0.4$ at 80% power requires about `ceiling(res$n) * 2` participants in total. A mixed model with 20 clusters of size 10 and a fixed effect of 0.3 has estimated power of `round(summary(ps)$mean * 100)`% in this simulation (50 replicates).

Common pitfalls

- Using the power formula for independent observations on a clustered design.
- Quoting an effect size taken from a pilot study without acknowledging its noise.
- Computing power from the smallest clinically important effect and the observed pilot effect interchangeably.
- Simulating with too few replicates.

Further reading

- Cohen J. *Statistical Power Analysis for the Behavioral Sciences.*
- Green P, MacLeod CJ (2016), *simr: an R package for power analysis of generalized linear mixed models.*
- Chow SC et al. *Sample Size Calculations in Clinical Research.*

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)